UNIT 10 At Sea - Changing the Watch

The Marina has successfully completed her loading and unloading in the Port of Antwerp. The weather is fine and the Marina is underway towards Scandinavia. She is passing along Öresund, with its great traffic density. Captain James is conning the ship. Mr Gray, the helmsman, is carrying out the orders.

- Full speed. Steer 150 degrees, orders Captain James.
- Steer 150 degrees, repeats Gray, the helmsman.

A few seconds later when the compass card settles on the mark, Gray says: "Steering 150 degrees, sir."

- Port 10 degrees.
- Port 10, sir.
- Ease to five, says the Captain and Gray brings the wheel back.
- Midships.
- Midships.
- Steer 130 degrees.
- Steering 130 degrees, answers Gray, as the heading steadies.

The Bridge

The navigational equipment on board the Marina includes satellite navigational equipment: a GPS (Global Positioning System) receiver, 2 radars (X-band with ARPA & S-band, AIS (Automatic Identification System), a gyro compass and a magnetic compass, a speed log, an electronic chart, an echo sounder and a wind measuring station.

The radio equipment includes the use of GMDSS (Global Maritime Distress and Safety System) for zones A1, A2 and A3 (A4). The GMDSS makes use of both satellite and conventional radio systems.

Sea Area A1 requires short range radio services - VHF is used to provide voice and automatic distress alerting via Digital Selective Calling (DSC).
Sea Area A2 requires medium range services. Medium Frequencies (MF - 2 MHz) are used for voice and DSC. Sea Areas A3 and A4 require long range alerting - High Frequencies (HF - 3 to 30 MHz), which are used for voice communication, DSC and Narrow Band Direct Printing (NBDP).

The Marina is equipped with a Navtex receiver and an Inmarsat ship station for navigational warnings, meteorological warnings, ice reports, search and rescue information, meteorological forecasts, pilot service messages, Satnav messages etc., and a 406 MHz EPIRB for emergency situations.

It is close to 12 o'clock and time to hand over the watch to the Second Mate Timo Ranta. He enters the bridge a few minutes early and is greeted by the Captain.

- Good morning Timo, says Captain James. Ready to take over?
- I am indeed. Any special problems?
- Well, we are in an area of high traffic density. The Öresund strait is not one of my favourites. It can be rather tricky. There is a great deal of crossing traffic and there seems to be extremely heavy traffic at the moment.
- Yes, that sounds familiar, I've been in these waters before.
- As you can see from the display our position is 56° 23´ North and 012° 01.0´ East, course 131 degrees speed 17 knots draught 7.0 metres. Passenger ferry is crossing from port side. The vessel should give way. Ro-ro vessel on our port bow, 1 mile ahead of us is on the same course. The bearing to the vessel is constant. Call me if a vessel passes with a CPA of less than 1 mile, and keep a 0.5 mile CPA for smaller boats all according to the standing orders. As you know there are a great number of fishing vessels in the area. So keep a sharp lookout - weather conditions seem to be changing. The wind is southwest 7 metres/second but visibility is expected to decrease, due to fog. The next weather report will be at 1200 UTC. You may have to reduce speed.
- Ok, understood.
- I switched to manual steering from autopilot at 1135.
- I see.
- GPS is working normally, Navtex is switched on and VHF DSC channel 70 and DSC controller switched on. Portside radar is at six miles range scale. Starboard radar is at twelve miles. The radars are at true motion presentation. The following requires special attention: Magnetic and gyro compass errors are as noted in the ship's logbook at the moment and the echo sounder is unreliable. At 0940 there was an engine alarm due to problems with the main engine. Speed was reduced to ten knots at 0945 and full speed resumed at 1010. Present revolutions of the main engine are, as you can see on the display, 116 per minute. There are no problems at the moment. There is no pumping of ballast tanks at the moment.

Deck crew is working on outside decks. Vessel is on an even keel. Last security patrol was an hour ago. Well, that about sums it up. You have the watch now.

- Ok, I have the watch, says Timo.
- Good luck. Call me if you are in doubt about anything, says Captain James as he leaves the bridge.

To navigate safely from one place to another, the ship's position must be recorded on charts. The positions must be checked very often. This can be done in different ways.

In the old days, when no land was in sight, observations of the sun, moon and stars were taken with a sextant and then the position was worked out. Satellite navigational equipment, the GPS, is today the most accurate and most important aid to navigation. Radar is used both on ships and ashore. Because the picture on the radar screen must not be disturbed by the ship's masts, funnels, derricks, etc. the radar scanner is usually installed very high.

In coastal areas it is possible to take bearings of the land or of seamarks by visual means or by radar. Today many ships have an electronic chart which shows the ship's position and route. Information on the route is fed into the ship's integrated navigational system and the ship can be switched from manual to automatic steering, the so-called auto-pilot.

If you need to find out how deep the water is, you can take soundings by using the echo-sounder, or a sonar. If it is cloudy, no land can be sighted, and all other navigational aids are out of function, the position may still be calculated by dead reckoning.

IMPORTANT! Study the following structures:

Do you remember these? Study these sentences, some of them from the text.

Radars **are used** both on land and at sea.
The radar scanner **is** usually **installed** very high.
The Marina **was built** in Finland.
Have the holds **been cleaned**?
The ship's position **must be recorded** on charts.